

Prac 7

#Practical 7

```
import nltk
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer

# Download necessary NLTK data
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')

# Sample Document
document = "The quick brown fox jumps over the lazy dog. Dogs are jumping and playing in the garden."
print("Original Document:", document)

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

# 1. Tokenization (Breaking into words)
tokens = word_tokenize(document.lower())

# 2. Stop words removal (Removing common words like 'the', 'is')
stop_words = set(stopwords.words('english'))
```

```
filtered_tokens = [w for w in tokens if w.isalnum() and w not in stop_words]
```

```
# 3. POS Tagging (Parts of Speech: Noun, Verb, etc.)
```

```
pos_tags = nltk.pos_tag(filtered_tokens)
```

```
# 4. Stemming (Chopping off suffixes)
```

```
ps = PorterStemmer()
```

```
stemmed = [ps.stem(w) for w in filtered_tokens]
```

```
# 5. Lemmatization (Finding the root dictionary word)
```

```
lemmatizer = WordNetLemmatizer()
```

```
lemmatized = [lemmatizer.lemmatize(w) for w in filtered_tokens]
```

```
print("Tokens:", tokens)
```

```
print("\nFiltered (No Stopwords):", filtered_tokens)
```

```
print("\nPOS Tags:", pos_tags)
```

```
print("\nStemmed:", stemmed)
```

```
print("\nLemmatized:", lemmatized)
```

```
# Create representation using TF-IDF
```

```
docs = [document] # TF-IDF usually takes a list of documents
```

```
vectorizer = TfidfVectorizer()
```

```
tfidf_matrix = vectorizer.fit_transform(docs)
```

```
# Convert to DataFrame for easy viewing
```

```
df_tfidf = pd.DataFrame(tfidf_matrix.toarray(),  
columns=vectorizer.get_feature_names_out())
```

```
print("\nTF-IDF Representation:")
```

```
print(df_tfidf)
```